MILITARY SPECIFICATION

INDICATOR, STERILIZATION

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of sensitized paper test record for determination of air elimination from within a sterilizer during the vacuum portion of a sterilization cycle.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive Water Resistant (For Sealing Waterproofed Paper)
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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Technical Association of the Pulp and Paper Industry (TAPPI) Standards and Suggested Methods.

TAPPI	T-411	Thickness (Caliper) of Paper and Paperboard
TAPPI	T-414	Internal Tearing Resistance of Paper
TAPPI	T-460	Air Resistance of Paper

(Application for copies should be addressed to Technical Association of the Pulp and Paper Industry, 360 Lexington Ave., New York, N. Y. 10017.)

3. REQUIREMENTS

3.1 Material. The sterilization indicator shall consist of a sensitized sheet of paper of white color on which is printed concentric circles of steam - heat - sensitive white pigment ink. The ink shall turn from white to black in the absence of air and exposure to steam heat during the vacuum portion of the sterilization cycle.

3.1.1 Physical properties. The indicator paper shall conform to the following properties when tested as specified in 4.4.1:

Basis weight, pounds per 500, 24 by 36.....38 ± 3
Thickness, inch..... .0043 ± .0005
Tearing resistance, grams, 16 sheets,
Weaker direction.....75 ± 17
Porosity, seconds, 100 ml. of air
through 1 sq in, 8 sheets.....30 ± 15

3.2 Description. Each sterilization indicator sheet shall be 9 inches by 12 inches and shall be sufficiently soft to conform to the irregular contours in surgical packs. The concentric circles of white ink shall begin with lines 7/16 ± 1/16 inch wide at the edge of the sheet and diminish in width and diameter toward the center of the sheet where there shall be a solid circle of ink 1 1/4 ± 1/16 inches in diameter. Once the ink has darkened, the chemical process shall be irreversible, providing a permanent record.

3.3 Performance. When subjected to a complete vacuum-steam sterilization cycle, at 260° F. and at 272° F. as specified in 4.4.2, the indicator's concentric circles of white ink shall turn black to indicate complete sterilization. When subjected to a sterilization cycle with partial vacuum and steam at 260° F. and at 272° F. as specified in 4.4.3, the ink color shall remain unchanged or shall darken only to gray to indicate incomplete sterilization.

3.4 Identification marking. The following information shall be printed on each indicator sheet:

Name or registered trademark of the manufacturer.
Product identification or tradename.
For record keeping purposes:

DATE
AUTOCLAVE NO. or STERILIZER NO.
OPERATOR

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The Federal Stock Number and Contract Number shall be located and marked in accordance with the PREPARATION FOR DELIVERY section of this specification (see 5.3.1). Instructions for use, and interpretation of results, shall accompany each unit of issue.

3.5 Instructions. Full instructions for use shall accompany each unit of 31 indicators.

3.6 Workmanship. Workmanship shall be first class throughout. Indicator sheets shall be free from defects which detract from their appearance or may impair their serviceability. Sheets shall be free of cuts, tears and holes, and shall be clean.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, covering physical properties (see 3.1.1) supplied by the manufacturer of the indicator may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements. The end item lots shall include material made from the same lot of paper and the same batch of ink.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I.

Table I.

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-1	1.0

Unit of product for examination shall be an individual sheet.

4.2.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105 and Table II."

Table II.

Characteristic **	Requirement	Test Procedure	Inspection level	AQL
Physical properties	3.1.1	4.4.1	S-1	*
Complete sterilization	3.3	4.4.2	S-1	*
Incomplete sterilization	3.3	4.4.3	S-1	*

* Acceptance number shall be zero (0) regardless of lot size.
** Unit of test for other than the physical properties tests shall be the individual paper sheet.

4.3 Examination. The indicators shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects, shown in Table III:

Table III.

Categories	Defects *
Major	
101	Indicating ink not applied on paper.
102	Indicating ink not applied in concentric cicles
103	Color of indication ink not as specified.
104	Paper sheet not free of cuts, tears, or holes
105	Paper sheet not clean.
Minor	
201	Identification marking missing, incomplete or illegible.

* Examination shall not be restricted to the classified possible defects listed above.

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4.4 Tests. Tests shall be conducted to determine compliance with all requirements contained in this specification.

4.4.1 Physical tests . Physical tests shall be conducted in accordance with this specification and applicable methods of TAPPI, as listed below:

Property	TAPPI Method
Thickness	T - 411
Tearing resistance	T - 414
Porosity	T - 460

4.4.2 Complete sterilization. A single sheet of the indicator shall be placed in the center of a stack of towels 10 to 11 inches high (2 to 3 dozen towels) and folded flat to 9 by 12 inches. The towels shall be wrapped as an ordinary surgical bundle and autoclaved in a pre-vacuum type sterilizer at 260° F. for 12 minutes. Repeat the test, using a fresh indicator, at 272° F. for 3 1/2 minutes. At the conclusion of the sterilization cycle, the indicators shall be examined.

4.4.3 Incomplete sterilization. The test shall be conducted at the same time and in the same sterilizer as the test in 4.4.2. A 9 inch strip cut from one side of the indicator paper shall be placed in a Mason jar, or equivalent tight seal container, in such a manner that half of the strip protrudes from the jar. The portion of the strip inside the jar shall be taped to the inner wall so that the temperature outside the jar reaches the strip while air is trapped in the jar. Before sealing, the jar and strip shall be heated in a 250° F. oven for 5 minutes in order to eliminate moisture from the sample and jar. The jar shall be tightly sealed, placed in the sterilizer and the two sterilization cycles completed as specified in 4.4.2. At the conclusion of each test, the test strip shall be examined.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package, containing 31 indicators and instruction sheet, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	48 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MIL-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B .

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. For use as a daily test record to determine if all air has been eliminated from within a tested sterilizer during the vacuum portion of a sterilization cycle, thereby attaining a sterilizing condition.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 This specification covers the following item appearing in the Federal Supply Catalog:

Stock Number	Item Identification
6530-181-7770	INDICATOR, STERILIZATION, SENSITIZED PAPER, 9 by 12 inches, 31s.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6530-1230

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

MIL-I-36789

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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